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Program 2: Develop a Multi-Stage Dockerfile for Container Orchestration

Project structure:

```
.
├── Dockerfile
├── index.html
├── package.json
├── package-lock.json
├── App.css
├── App.jsx
├── assets
│   └── react.svg
├── index.css
└── main.jsx
```

STEP 1: Create a Dockerfile and add the following content.

```
#build stage
FROM node:20-alpine AS builder
WORKDIR /app
COPY package*.json ./
RUN npm i
COPY . .
RUN npm run build

#preview stage
FROM node:20-alpine AS test
WORKDIR /app
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/package*.json ./
RUN npm i
EXPOSE 4173
CMD ["npx", "vite", "preview", "--host", "0.0.0.0", "--port", "4173"]
```

STEP 2: Execute the Docker Build command on the Terminal to build the Docker image and then execute the Docker run command specifying the Port Number for the container.

➤ **sudo docker build --target builder -t react-builder .**

```
> sudo docker build --target builder -t react-builder .
[+] Building 1.2s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 407B                                0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  1.1s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281 0.0s
=> [internal] load build context                                  0.1s
=> => transferring context: 266.46kB                                0.1s
=> CACHED [builder 2/6] WORKDIR /app                              0.0s
=> CACHED [builder 3/6] COPY package*.json ./                     0.0s
=> CACHED [builder 4/6] RUN npm i                                  0.0s
=> CACHED [builder 5/6] COPY . .                                  0.0s
=> CACHED [builder 6/6] RUN npm run build                         0.0s
=> exporting to image                                             0.0s
```

› sudo docker build --target test -t react-test .

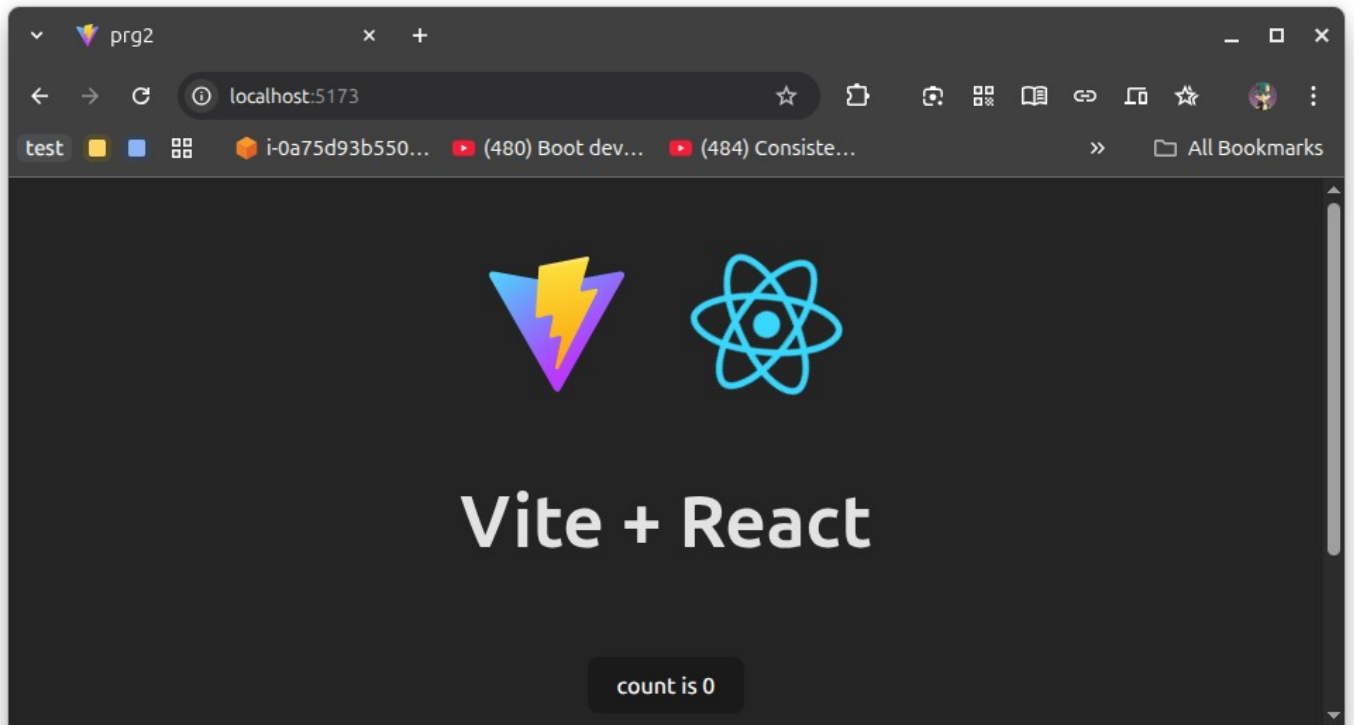
```
> sudo docker build --target test -t react-builder .
[+] Building 1.3s (14/14) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 407B                                0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  1.1s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                      0.0s
=> [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281 0.0s
=> [internal] load build context                                  0.1s
=> => transferring context: 266.46kB                               0.1s
=> CACHED [builder 2/6] WORKDIR /app                              0.0s
=> CACHED [builder 3/6] COPY package*.json ./                     0.0s
=> CACHED [builder 4/6] RUN npm i                                  0.0s
=> CACHED [builder 5/6] COPY . .                                   0.0s
=> CACHED [builder 6/6] RUN npm run build                         0.0s
=> CACHED [test 3/5] COPY --from=builder /app/dist ./dist        0.0s
=> CACHED [test 4/5] COPY --from=builder /app/package*.json ./   0.0s
=> CACHED [test 5/5] RUN npm i                                    0.0s
=> exporting to image                                             0.0s
=> => exporting layers                                             0.0s
=> => writing image sha256:b8a818310dbd971ae5d261adcac59f2ab5992219e5c9631cb1fa8f5c74 0.0s
=> => naming to docker.io/library/react-builder                  0.0s
```

STEP 3: you will get two images one for development and other for test.

```
> sudo docker images
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
react-builder:latest 247b7fc12d16 253MB 0B
react-test:latest b8a818310dbd 202MB 0B
```

STEP 4: run the containers.

```
> sudo docker run -p 5173:5173 react-builder
[sudo] password for kumars:
> sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
> sudo docker images
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
react-builder:latest 247b7fc12d16 253MB 0B
react-test:latest b8a818310dbd 202MB 0B
> sudo docker run -p 5173:5173 247b7fc12d16
> sudo docker run -p 5173:4173 react-test
→ Local: http://localhost:4173/
→ Network: http://172.18.0.2:4173/
```



STEP 4: Stop and remove the container with the container number, remove the docker container and delete the Docker Image

› **sudo docker rmi react-builder**

› **sudo docker rmi react-test**